

CTBI Normalization Audit

Independent check of Dev's min-max normalization and CTBI recurrence implementation, run against the real CTBI.csv output.

A. Min-max normalization bounds

- DayStress_norm stays within [0, 1]: ✓
- NCTL_norm stays within [0, 1]: ✓
- Formula used: $(x - \min(x)) / (\max(x) - \min(x))$ — standard min-max, correctly implemented, with an explicit divide-by-zero guard (`if max == min: return 0.0`) for degenerate series.

B. Normalization actually spans the full [0,1] range

- DayStress_norm hits exactly 0 and 1 at series extremes: ✓
- NCTL_norm hits exactly 0 and 1 at series extremes: ✓

C. CTBI weight-sum check ($\alpha + \beta + \gamma$) across the sensitivity test

α tested	β	γ	$\alpha+\beta+\gamma$	Note
0.6	0.15	0.15	0.90	Weights sum < 1 — CTBI will trend slightly lower at steady state
0.7	0.15	0.15	1.00	Balanced — CTBI stays same order of magnitude at steady state
0.8	0.15	0.15	1.10	Weights sum > 1 — CTBI will trend slightly higher / can grow unbounded over many days

Finding: Only $\alpha = 0.70$ makes the recurrence weights sum to exactly 1.0 (a true weighted average of CTBI($t-1$), DayStress_norm, NCTL_norm). At $\alpha = 0.60$ the weights sum to 0.90 (slight downward drift each day), and at $\alpha = 0.80$ they sum to 1.10 (slight upward drift each day). This does **not** invalidate the sensitivity test — the playbook only asks that the *qualitative pattern* (rise-to-peak, decline-after) stay stable across α , which it does

(confirmed independently by Khushi's Day 3 sanity check, correlation of NCTL $\leftrightarrow$ peak-CTBI = 0.944). But it is worth one line in the Decision Log so nobody claims $\alpha=0.6/0.8$ CTBI values are on the exact same scale as $\alpha=0.7$ — they are close, not identical, by construction.

D. CTBI output range sanity ($\alpha = 0.70$, the locked default)

- CTBI min: 0.094
- CTBI max: 0.768
- Since DayStress_norm and NCTL_norm are both bounded to $[0,1]$, and $\alpha+\beta+\gamma = 1.0$ exactly at $\alpha=0.70$, CTBI is mathematically bounded within $[0, 1]$ at steady state for the default α . Observed range above confirms this — no runaway or negative values.



Overall: ☒ Normalization implementation is correct

One documentation note recommended for the Decision Log (not a bug): the $\alpha+\beta+\gamma$ weight-sum only equals exactly 1.0 at $\alpha=0.70$; the 0.60/0.80 sensitivity runs are intentionally slightly off-balance, per the playbook's own sensitivity-test design.